

IB Economics – internal assessment coversheet

Section 1

Title of Article: Air pollution takes 2 years off average global life expectancy, more than smoking or alcohol

Source of Article:

<https://www.cnbc.com/2022/06/14/air-pollution-takes-2-years-off-your-life-more-than-smoking-or-alcohol.html?qsearchterm=pollution>

Date the Article was Published: 2022-06-14

Date the Commentary was Written: 2022-10-08

Word Count: 785

Key Concept: Sustainability

Section of the Syllabus the Article Relates to: Microeconomics

Section 1

Air pollution takes 2 years off average global life expectancy, more than smoking or alcohol

Air pollution, which is primarily the result of burning fossil fuels, takes 2.2 years off the average global life expectancy, according to a new report out Tuesday from the Energy Policy Institute at the University of Chicago (EPIC).

The Air Quality Life Index, or AQLI, finds that taken together, air pollution takes a collective 17 billion years of life, and reducing air pollution to meet international health guidelines would increase the global average life expectancy from roughly 72 to 74.2 years.

Firsthand cigarette smoke reduces life expectancy by 1.9 years, on average, according to the report. Alcohol and drug use reduce life expectancy by nine months on average, unsafe water and sanitation reduce expectancy by seven months, HIV and AIDS reduce life expectancy by four months, malaria reduces average life by three months, and conflict and terrorism reduce life expectancy by seven days, the report said.

The AQLI report is notable because its estimate of the impact on particulate pollution on human life expectancy is based on research that allows it to show causation, not just correlation. “Because of the way these studies were designed - and the quite fortuitous set of policies that enabled that design, they established a causal, rather than a correlative, relationship between particulate matter exposure and mortality,” Christa Hasenkopf, the director of AQLI, told CNBC.

Air pollution is so dangerous because it is impossible to avoid, especially for people who live in particularly polluted locations, the report says. “Whereas it is possible to quit smoking or take precautions against diseases, everyone must breathe air. Thus, air pollution affects many more people than any of these other conditions,” the report says. Sixty percent of particulate matter air pollution is caused by fossil fuel combustion, 18% comes from natural sources (including dust, sea salt, and wildfires), and 22% comes from other human activities.

The report, developed by the University of Chicago’s Michael Greenstone and his team at the EPIC, is a measurement of the air pollution in 2020, when the Covid-19 pandemic was reducing activity and transportation.

The massive contraction of activity reduced global pollution levels only by a tiny bit. Population weighted-average particulate matter declined from 27.7 micrograms (one-millionth of a gram) per cubic meter of air to 27.5 micrograms per cubic meter of air between 2019 and 2020, according to the report.

And in South Asia, where air pollution is the most dire, the air pollution rose in 2020 from the year prior. India, Pakistan, Bangladesh and Nepal are among the most polluted countries in the world.

Particulate matter air pollution is suspended in the air and categorized by its size. The smaller it is, the deeper it can get into the body. Particulate matter with a diameter of less than 10 micrometers, often designated PM10, can pass through the hairs in the nose,

down the respiratory tract and into the lungs.

Smaller particulate matter with a diameter less than 2.5 micrometers, often designated as PM_{2.5}, is about 3% the diameter of a human hair and can get into the bloodstream by way of the lungs' alveoli. It can affect blood flow, eventually causing a stroke, heart attack and other health issues.

When the World Health Organization first published air quality guidance in 2005, it said the acceptable levels of air pollution was less than 10 micrograms per cubic meter. In September, the World Health Organization changed its benchmark guidelines to below 5 micrograms per cubic meter.

Currently, 97.3% of the global population, equaling 7.4 billion people, live in places where the air quality does not pass the recommended 5 micrograms per cubic meter limit recommended by the WHO for particulate matter with a diameter of less than 2.5 micrometers.

"This report reaffirms that particulate pollution is the greatest global health threat," wrote Greenstone, who was previously the chief economist for former President Barack Obama's Council of Economic Advisers. "Yet we also see the opportunity for progress. Air pollution is a winnable challenge. It just requires effective policies."

For example, China has been able to dramatically improve its air quality. In 2014, after a year in which China had record levels of pollution, the then premier, Li Keqiang, declared a "war against pollution." The government spent money to combat pollution and was able to reduce particulate pollution by 39.6%, the report says.

Despite China's progress, the air pollution levels in China are still more than what the WHO recommends.

"It is important to note that air pollution is also deeply intertwined with climate change. Both challenges are primarily caused by the same culprit: fossil fuel emissions from power plants, vehicles, and other industrial sources," the report's executive summary says. "These challenges also present a rare win-win opportunity, because policy can simultaneously reduce dependence on fossil fuels that will allow people to live longer and healthier lives and reduce the costs of climate change."

The American Medical Association, the country's largest physician trade group, voted on Monday to adopt a policy to declare climate change a public health crisis.

"The scientific evidence is clear — our patients are already facing adverse health effects associated with climate change, from heat-related injuries, vector-borne diseases and air pollution from wildfires, to worsening seasonal allergies and storm-related illness and injuries. Like the COVID-19 pandemic, the climate crisis will disproportionately impact the health of historically marginalized communities," said AMA Board Member Ilse R. Levin, in a written statement announcing the vote. "Taking action now won't reverse all of the harm done, but it will help prevent further damage to our planet and our patients' health and well-being."

Commentary

According to the report from Energy Policy Institute at the University of Chicago (EPIC), the air pollution has a serious damage to people's health and will reduce the life expectancy as well. The World Health Organization then published a policy and suggested a possible maximum cap for the amount of air pollution per cubic meter. This action taken is related to the concept of **sustainability** which refers to the long-term maintenance or viability of any particular activity or policy and to maintain people's healthy living conditions.

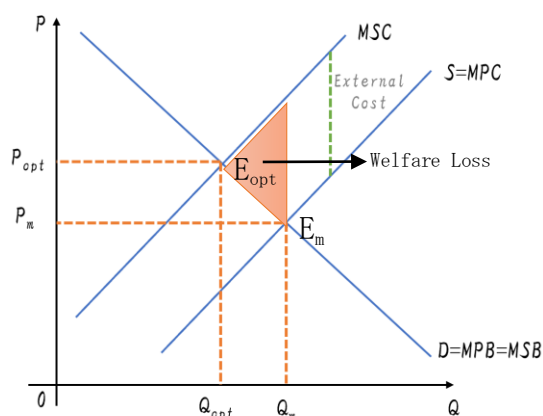


FIGURE 1: Air Pollution arose from Production.

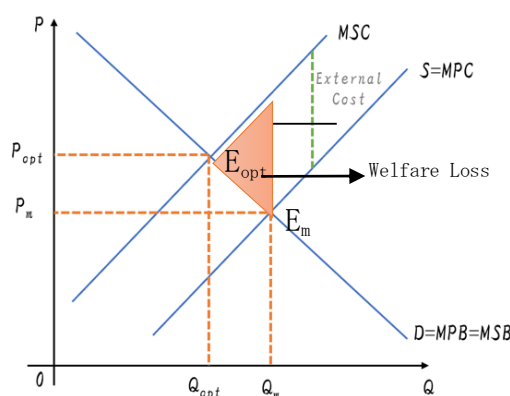


FIGURE 2: After the Intervention

“Air pollution, which is primarily the result of burning fossil fuels, takes 2.2 years off the average global life expectancy.” This means that burning fossil fuels will have externalities on people. **Externality** is defined as when producing or consumption of a product will have side effects on third parties whose interests are not considered. In this case, burning fossil fuels will lead to pollutions especially in air and lead to negative effect to people who are not considered in the production or using of fossil fuels. This matches the definition for negative externality, especially matched the concept of **Negative Production Externality** which means the side-effects on third parties involve costs arise by producing of a good.

To begin with, in the market where the production will need to burn fossil fuels, the demand curve and supply curve intersects at E_m , the Marginal Private Benefit (MPB) is equal to Marginal Private Cost (MPC) and the market is at the equilibrium. However, the burning of fossil fuels will lead to emission of Carbon Dioxide and other harmful gases which will both do bad to people and the environment. This will lead to more cost when dealing with this kind of side effects. Therefore, the Marginal Social Cost (MSC) in order to fix this problem will be larger than the cost during production. It indicates that MSC is lying on up of MPC, which is shown in FIGURE 1. MSC and MPB (also is MSB) will then intersect at point E_{opt} with higher price and lower quantity of goods and services produced. The distance between MSC and MPC means the external cost in order to deal with the negative side effects. In the article, “in September, the World Health Organization changed its benchmark guidelines to below 5 micrograms per cubic meter”, by doing it, it will limit the amount of pollutants released and so to protect the health of people which relates to the concept of **sustainability**. All the countries in the organization (WHO) will follow it, and by reducing the emission, they may decrease their producing process so to reduce the emission of harmful gases from burning fossil fuels. This will decrease the supply curve of producing and shift the supply curve to the

left. In this case, it will shift MPC curve to the left to approach the MSC curve. The market equilibrium will move from E_m to E_{opt} with decreasing in quantities produced and increase in the price. In reality, E_m will not directly move to the level of E_{opt} , it will only tend to approach it.

Moreover, this policy can have some advantages in non-market areas. In the article, it says that “these challenges also present a rare win-win opportunity, because policy can simultaneously reduce dependence on fossil fuels that will allow people to live longer and healthier lives and reduce the costs of climate change”. By doing this policy, more people can benefit from it since it will increase people’s life expectancy greatly and will increase the living standard of people in the country. This reflects the topic of **sustainability**. Moreover, less cost will be used to deal with climate problems. Since less fossil fuel is burned and less harmful gases are released, the environment will become better and more input can be used on other fields rather than climate problems. By the way, this will also increase the living standard of people.

On the other hand, such policy will also have some bad effects on the countries taking it. The policy is about setting a maximum number of pollutants in the air to solve the problem of air polluting. However, the amounts of pollutants restricted is hard to decide. If the amounts of pollutants are too big, it will have nearly no effect on the protection of environment; if the amounts of pollution is set too some, it will lead to political dissatisfactions. Because, it is impossible to release no pollution during production nowadays, a too restricted policy may cause difficulties for some businesses to continue producing. This will lead to disagreement from those businesses and will eliminate the effect of the policy.

To conclude, the **sustainability** not only promote economics growth but can also help maintain the living standard of people and the standard of nature as well. However, if the policy is not controlled correctly, for instance, too restrict or too loose, the effect will be eliminate.

Section 2

Title of Article: UK announces govt spending cuts, tax hikes

Source of Article:

<https://global.chinadaily.com.cn/a/202211/18/WS6376dca3a31049175432a7b1.html>

Date the Article was Published: 2022-11-18

Date the Commentary was Written: 2023-01-22

Word Count: 800

Key Concept: Intervention

Section of the Syllabus the Article Relates to: Macroeconomics

Section 2

UK announces govt spending cuts, tax hikes

Hot on the heels of the United Kingdom's inflation rate hitting a 41-year high of 11.1 percent, the nation's finance minister, Chancellor of the Exchequer Jeremy Hunt, unveiled a budget on Thursday aimed at restoring stability.

Hunt's budget, officially termed his autumn statement, called for around 30 billion pounds (\$35.6 billion) of government spending cuts and 24 billion pounds of additional taxes and was aimed, he said, at closing a spending gap in the government's accounts and avoiding additional national debt.

Hunt told lawmakers the UK has been dealing with "unprecedented global headwinds" and, with the cost-of-living crisis continuing, "families, pensioners, businesses, teachers, nurses, and many others "are concerned about the future.

"So, today, we deliver a plan to tackle the cost-of-living crisis and rebuild our economy," he said.

Hunt, who replaced previous chancellor Kwasi Kwarteng after his disastrous mini-budget in September, said several things have hurt the economy, including the high cost of energy triggered by shortages attributed to the Russia-Ukraine conflict, and the fight against the novel coronavirus pandemic. But he insisted, despite official forecasts that the economy will shrink by 1.4 percent next year and news the nation is in recession, that the government will "protect the vulnerable" because "to be British is to be compassionate and this is a compassionate government".

He said his budget was also aimed at promoting "stability, growth, and public services".

Included in the budget was a freeze on income tax thresholds, which, because of fast-rising wages triggered by high inflation, means millions more people will pay additional income tax. Hunt said the change means the UK's income tax burden as a proportion of national income will go up by around 1 percent during the next five years.

The chancellor also lowered the threshold at which the UK's highest earners must pay tax at the 45 percent rate, from 150,000 pounds to 125,140 pounds a year.

He also said he will collect more tax from energy companies, which have enjoyed record turnovers because of the high cost of fuel, with the UK's windfall tax set to rise from 25 percent to 35 percent. He said the UK will also save money by cutting the amount it spends on overseas aid, with it lowered from 0.7 percent of GDP to 0.5 percent.

The government will also reduce the support it gives households for energy bills, which are set to rise in April from 2,500 pounds to 3,000 pounds a year as a result.

But there was some better news for the nation's poorest people, with state pensions, welfare payments, and tax credits all set to rise by 10.1 percent, and with the minimum legal wage set to go up from 9.5 pounds an hour to 10.42 pounds.

Some members of Parliament from Hunt's own Conservative Party criticized the budget, saying the party should be a party of tax cuts, not hikes.

Esther McVey, a former Cabinet minister, said tax rises were the "last thing" the party needed, while lawmaker Simon Clarke, who was in former prime minister Liz Truss's Cabinet, said he feared Hunt had "overcorrected" Kwarteng's mistakes.

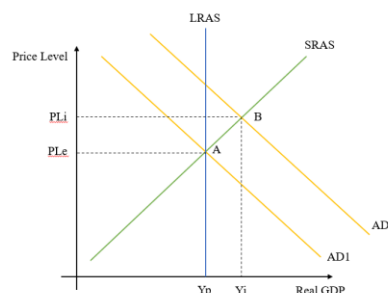
Rachel Reeves, the opposition Labour Party's shadow chancellor, said Britain should be unveiling a "proper plan for growth" instead of announcing tax hikes.

"The chancellor has deployed a raft of stealth taxes, taking billions of pounds from working people," she told lawmakers. She said the combination of high inflation and tax hikes will cost the average person 600 pounds a year.

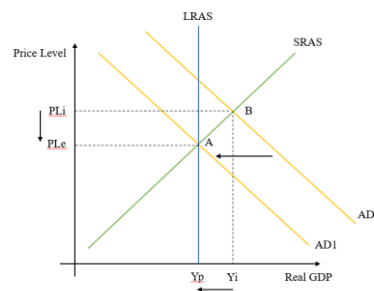
Commentary

This article is about the ways that UK government took in order to reduce the inflation that happened during that time. This is a big inflation, in the article, it says that “The United Kingdom's inflation rate hitting a 41-year high of 11.1 percent.”. The rate of inflation reached 11% which means it would have a serious effect on the economy. The UK government had intervened in the economy which matches the key concept of **intervention**. It typically refers to government intervention, meaning that the government becomes involved with the workings of markets.

According to the article, UK economy is experiencing an inflation, in the article it says that “inflation rate hitting a 41-year high of 11.1 percent”. The inflation rate indicates changes in the general price level. In this case, the inflation rate is high which states an increasing in price level. Moreover, it is a generally severe inflation since usually economists expects an inflation rate below 5 percent. In the article, it mentions that “the cost-of-living crisis continuing”. It indicates that there is an increase in cost of living for households and firms. The high cost of living which are factors of aggregate demand (AD) and will shift AD curve to the right, illustrated by the Neo-Classical Model below:



Currently, UK economy is at point B with real GDP Y_i (Real GDP during inflation) and price level PL_i (Price Level during inflation). An inflationary gap occurs at $Y_i - Y_p$. This shows an increasing in price level, an increasing in real GDP and less unemployment rate compared to point A which is short-run equilibrium.

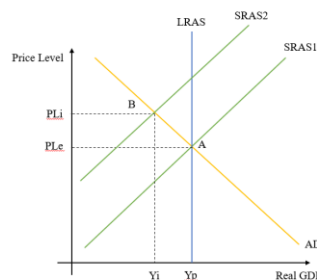


According to the article, “Hunt's budget, officially termed his autumn statement, called for around 30 billion pounds (\$35.6 billion) of government spending cuts and 24 billion pounds of additional taxes and was aimed.” This strategy will decrease government spending and increase taxes as well. These policies are all fiscal policies which is manipulations by the government of its own expenditures and taxes in order to influence the level of aggregate demand. By decreasing government spending, less money will be used for firms to do investment. This will decrease aggregate demand (AD) and shift it to the left. Government spending has a direct impact on the level of aggregate demand. Meanwhile, by raising taxes like income and business tax, it will decrease the consumption power of consumers and thus decrease consumer spendings which is a component of AD. These two aspects will all decrease AD and shift it to the left. It will

decrease real GDP from Y_i to Y_p , decrease price level from PL_i to PL_e . It will also increase unemployment rate.

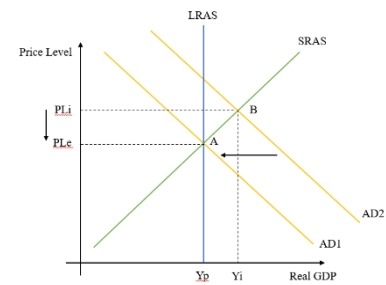
Not as same as in the theory, the real GDP will not reach the potential output exactly. This may because of fluctuations of price level and changes in the economic situation rapidly with time. Therefore, an only policy will not solve economic problems completely. Potential output can only be reached in the theory.

In the article, it mentions some potential reasons for such economy crisis to take place. For example, “the high cost of energy triggered by shortages attributed to the Russia-Ukraine conflict” and “the fight against the novel coronavirus pandemic”. In the graph it can be shown:



Because of high energy cost, it will increase the cost of production of firms. This will affect SRAS directly by shifting it to the left. Moreover, the COVID pandemic can be considered as a supply shock which is negative. This will also shift the curve to the left. Therefore, price level will increase in these situation from PL_e to PL_i . This is also a sign of inflation.

According to the policy, the government decreases their spendings and increases the taxes for households and firms. By decreasing government spendings and increasing taxes, it will shift AD to the left. This is because the disposable income for firms and households, they will decrease their spendings which is components of AD. In the graph, it will shift the AD curve from right to the left, from point B to point A. Price level will decrease from PL_i to PL_e , real output from Y_i to Y_e . In the real world, the real output will not reach the potential but will only eliminate inflationary gap.



However, there is also some people that don't agree with the policy, especially the raising taxes. According to the article, “Some members of Parliament from Hunt's own Conservative Party criticized the budget, saying the party should be a party of tax cuts, not hikes.” This indicates that the policy like raising taxes will have political constraints which is not good for the implement of the policy. The reason is that “The chancellor has deployed a raft of stealth taxes, taking billions of pounds from working people” and they think it's better to establish plans for growth of economy instead of just raising taxes. These policies may affect the economy in the long run, such as increasing technologies or increase the quality and quantity of labors. This may include establish better education and health care systems.

Section 3

Title of Article: EU carbon tax and other laws to impact \$37 billion of India's exports:
Report

Source of Article:

<https://economictimes.indiatimes.com/news/economy/foreign-trade/eu-cbam-other-laws-to-impact-37-billion-of-indias-exports/articleshow/103836767.cms>

Date the Article was Published: 2023-09-21

Date the Commentary was Written: 2023-10-15

Word Count: 781

Key Concept: Equity

Section of the Syllabus the Article Relates to: The Global Economy

Section 3

EU carbon tax and other laws to impact \$37 billion of India's exports: Report

New Delhi: The Council on Energy, Environment and Water has said that India's exports of around \$37 billion, which is approximately 43% of the country's exports to the EU as of 2022, are likely to be impacted due to the bloc's various non-tariff measures (NTM) including the Carbon Border Adjustment Mechanism (CBAM).

Some of the recent regulations that can prove to be NTMs include the EU's CBAM and Deforestation-free Regulation (EUDR), and the United States' Inflation Reduction Act, it said. NTMs are policy measures other than ordinary customs tariffs that can have an economic effect on trade.

"While developed countries have lowered their tariff levels over time, they have gradually increased the issuance of NTMs," the think tank said.

As per the report titled 'Sustainability-driven Non-Tariff Measures: Assessing Risks to India's Foreign Trade', the product categories at risk due to the proposed sustainability-focused EU regulations include textiles, chemicals, selected consumer electronics products, plastics, and vehicles.

These items accounted for 32% of India's exports to the EU in 2022, valued at approximately \$27 billion.

It noted that regulatory differences between trading partners make compliance with these complex measures difficult for countries without equivalent regulations, leading to restrictions in market access, specifically for emerging economies such as India.

The think tank has suggested trade pacts, stronger compliance through common information-sharing platform for firms to register and the information can be tracked, and introduction of India's own standards and non-tariff measures to ensure the quality of its manufactured items.

Sustainability-driven NTMs are increasing at a high rate and their proportion of the total NTM notifications to WTO has increased to 19% in 2021 from 8% in 1997.

NTMs implemented by the EU have impacted India's rice and chemical exports in the past, according to the report.

"With an increase in the issuance of E-NTMs by developed countries, India faces a serious challenge concerning its key export items," the center said.

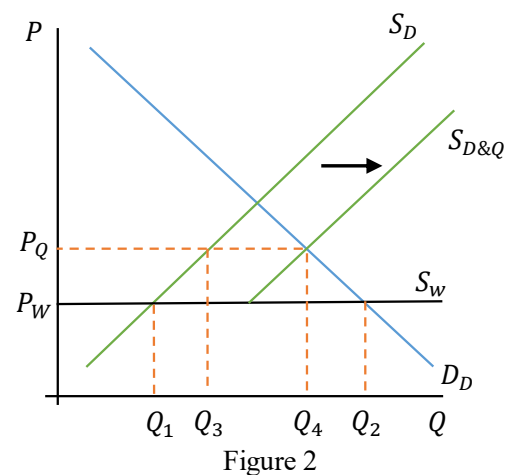
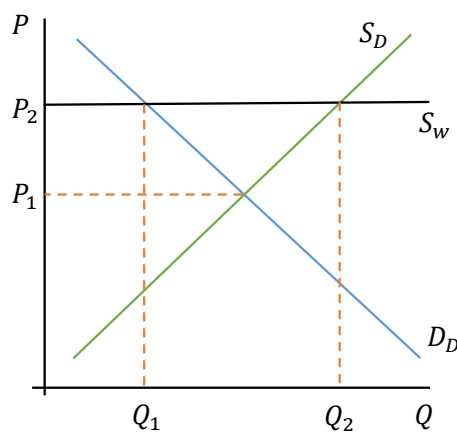
It has suggested a "structured approach" to deal with these measures to ensure that India's exports are not impacted. It includes developing mutual recognition of compliance assessment activities in the respective countries through bilateral trade pacts, raising specific trade concerns, preparing the industry.

Commentary

In the article, it mentions that India is facing an issue based on the restriction of its exports to the EU. In the article, it states that “these items accounted for 32% of India’s exports to the EU in 2022, valued at approximately \$27 billion”. Because of the Non-Tariff measures apposed by the EU, there is a huge impact on the export revenue of India. According to the article, the part of industry that is affected in India is the exports of chemicals and rice. It fulfills the requirement of economic concept **equity** because in the article it says an “introduction of India’s own standards and non-tariff measures to ensure the quality of its manufactured items” in order to overcome the restrictions result by the policy of EU.

Firstly, about the policy that is introduced by the EU. The Non-Tariff Measures, in short NTMs, are policies that measure other than ordinary customs tariff that can have an economic effect on trade. This means that the restrictions are not provided by imposing of tariff but rather by some other factors that directly effect the supply or demand of imports from other countries. This includes Carbon Border Adjustment Mechanism, in short CBAM, imposed by the EU. CBAM policy is one of the policies in order to protect the sustainability of development. In the article, India cannot fulfill the requirement of CBAM and thus being restricted by the EU.

Having a look at the situation of India economy, it is providing exports to the EU. Therefore, its graph is labelled as exporters. In Figure 1, the price of the world supply is at P_2 which is higher than the domestic price which is at P_1 . The quantity supplied domestically is larger than the demand domestically ($Q_2 > Q_1$). Therefore, the quantity exported is $Q_2 - Q_1$ for India.



In the article, the EU is imposing NTMs to the exports of India. This limits the quantity imported to the EU; hence, it can be considered similar in the function of quota. Quota is a type of trade protection that involves setting a legal limit to the quantity of a good that can be imported over a particular time period. The effect on the EU economy is illustrated through Figure 2 above. In order to limit the import, a new supply curve $S_{D\&Q}$ is created and eliminate the quantity import from $Q_2 - Q_1$ to $Q_4 - Q_3$. The price of the imports still remains at P_W . Consider the India economy on the other hand, it is still selling its products at world price P_W . However, the amounts of exports it is

exporting decreases. This leads to the loss of its export revenue. In the article, it mentions that “India’s exports of around \$37 billion, which is approximately 43% of the country’s exports to the EU as of 2022”, saying that there will be a very huge impact on the economy of India if the EU imposed the NTMs to India.

In the article, it suggests that India can introduce its “own standards and non-tariff measures to ensure the quality of its manufactured items”. This is a form of retaliation which is a policy that restrict the country or organizations that impose restrictions to the home country. By doing such policy, India can also limit the imports from the EU. Less imports are imported into India which means that India can have the ability to control its amount of domestic manufactured goods. In the passage, it mentions that the country has the ability “to ensure the quality of its manufactured items”. Although this method can protect the domestic products, it will make the resources allocate less efficiently. Since the world price is lower than domestic price and India is an exported country, the domestic price will be very high. These products are produced by less efficiently domestic producers which leads to the misallocation of resources.

In conclusion, in this news article, it mentions that India is experiencing an NTM policy from the EU. India is an export country since its amount of domestic products supplied is larger than its domestically demanded. Moreover, its domestic price is larger than world price, it is a better choice for India to export its products. However, due to NTMs, the amount of exports from India is strictly reduced. Since these exports are essential for the India economy, India established a retaliation. This policy has the ability to protect domestic producers from the impact of foreign producers. However, this will make the misallocation of resources since more products are produced by inefficient domestic producers.